St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank (Revised Curriculum 2020 onwards)

First Year- First Semester (2020 -21 Batch)

Course Title (Paper Title): GREEN CHEMISTRY QP Code: 85151

Unit/ Module	SI. No	Questions	Marks
1	1.	Explain Real-time analysis for pollution prevention	2
1	2.	Mention two traditionally used Halogenated and Hydrocarbon solvents.	2
1	3.	How ethyl L-lactate is synthesized?	2
1	4.	Explain E-factor.	2
1	5.	Differentiate Percentage yield from Atom economy.	2
1	6.	Mention any two greener energy input methods.	2
1	7.	List advantages of carbohydrates as green starting materials.	2
1	8.	How Kojic acid is prepared from glucose?	2
1	9.	How Furfural is prepared from carbohydrate source?	2
1	10.	What are the application of Polylactic acid?	2
1	11.	How Polylatic acid is prepared?	2
1	12.	Mention any four green reagents.	2
1	13.	Enlist the advantages of Polymer supported reagents.	2
1	14.	Mention advantages of water as green solvent.	2
1	15.	Indicate the uses of solvents in organic synthesis.	2
1	16.	Mention the advantages of biocatalytic conversions in relevance to green chemistry.	2
1	17.	Enlist the various class of green catalyst.	2
1	18.	What are biocatalyst? Give an example.	2
1	19.	What are PTC? Give an example	2
1	20.	Define catalyst with an example.	2
1	21.	What are binary ionic liquids? Give an example.	2

1	22.	Mention any two green Acid catalyst.	2
1	23.	Mention any two Oxidation catalyst with greener approach.	2
1	24.	Mention any four class of enzymes.	2
1	25.	Why microwave heating is described as an "**instant on- instant off"?**	2
1	26.	Mention the advantages of microwave exposure.	2
1	27.	Differentiate conventional heating and microwave heating.	2
1	28.	Illustrate neat reaction	2
1	29.	What are solid support reactions? Give an example.	2
1	30.	Give the structures of linear and branched chain alkylbenzenesulphonates.	2
1	31.	Write green synthesis protocol for synthesis of Adipic acid.	2
1	32.	Mention four green reagents.	2
1	33.	What are Photocatalyst? Give an example.	2
1	34.	What are polymer supported catalyst? Give example.	2
1	35.	Give an example for mw assisted condensation reaction.	2
1	36.	Give an example for mw assisted oxidation reaction.	2
1	37.	Give an example for mw assisted reducttion reaction.	2
1	38.	Give an example for mw assisted functional group transformation reaction.	2
1	39.	Mention the two solvent less approach employed in microwave synthesis.	2
1	40.	Write an example for mw assisted Ugi reaction.	2
1	41.	What is the frequency of microwaves used in industrial oven and RADAR/telecommunications?	2
2	42.	What are ionic liquids? Give an example	2
2	43.	Why molten NaCl does not consider as ionic liquids?	2
2	44.	What are ionic liquids? Also write its other names.	2
2	45.	Give any two example of ionic liquids cation	2
2	46.	Give any two example of ionic liquids Anion	2
2	47.	Write any two structure of ionic liquids	2
2	48.	Write some common used Room temperature Ionic liquids	2
2	49.	Mention any four physical properties of ionic liquids	2
2	50.	Why ionic liquids are so attractive?	2
2	51.	An ionic liquid is an excellent solvent. Justify	2

2	52.	Why ionic liquid exists in liquid state at room temperature?	2
2	53.	Ionic liquids are "liquids". Give reasons.	2
2	54.	Write the structure of Ethyl-3-methylimidazolium Chloride	2
2	55.	Write the structure of 1-Butyl-3-methylimidazolium hexafluorophophate	2
2	56.	Write the structure of 1-butyl-3,5-dimethylpyridinium bromide	2
2	57.	Why ionic liquids consider as Green Solvents.	2
2	58.	Give the name of first room temperature ionic liquids with Melting point.	2
2	59.	What is alkylation? Name one ionic liquids used in this reaction.	2
2	60.	Name the pyridinium ionic liquid used in alkylation process.	2
2	61.	Name any two ionic liquids used in alkylation process.	2
2	62.	Name the conversion process used in petroleum refineries. Mention it use.	2
2	63.	What is Hydroformylation reaction? Hence also write its other name.	2
2	64.	Name the industrial process used for the production of aldehyde. Mention the starting material in this process.	2
2	65.	Define Hydroformylation reaction. Mention the catalyst used in this process.	2
2	66.	What is Epoxidation? Write its equation	2
2	67.	Write the structure of Katsuki-type catalyst.	2
2	68.	How Katsuki-type catalyst is good than Jacobsen-type catalyst.	2
2	69.	Write the structure of [Bmim]BF_4	2
2	70.	What is Friedel-Crafts process? Hence, write the ionic liquid used in Friedel Crafts Alkylation.	2
2	71.	What is used of ionic liquid in the Friedel-Crafts alkyaltionProcess.	2
2	72.	Write any two ionic liquids used in Friedel-Crafts alkyaltion.	2
2	73.	What is Diels-Alder reactions? Mention one ionic liquids used in this reaction.	2
2	74.	Write the neutral ionic liquids used in Diels-Alder reactions.	2
2	75.	Why ionic liquid is used in Diels-Alder reactions?	2
2	76.	Write any two ionic liquids used in Diels-Alder reactions.	2
2	77.	What is Knoevengal condensation reaction? Mention its one application	2
2	78.	What is Knoevengal condensation reaction? Write one ionic liquids used in this reaction	2

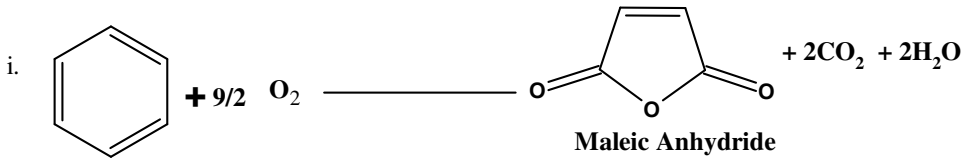
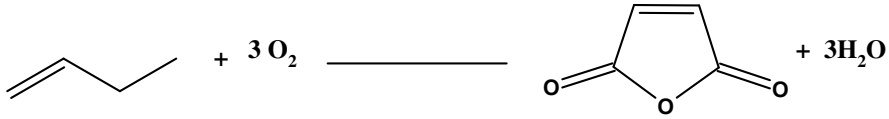
2	79.	Why Ionic liquids are used in Knoevengal condensation reaction?	2
2	80.	What is Advantage of ionic liquids in Knoevengal condensation reaction?	2
2	81.	Ionic liquids of amino acid amides used in Knoevengal condensation reaction. Give reasons	2
2	82.	Write the structure of any ionic liquids used in Knoevengal condensation reaction.	2
2	83.	Write the structure of (1-((4-Cholrophenyl)amine)-1-oxo-propan-2-aminium per chlorate.	2
2	84.	What is a Wittig reaction? Mention its one application	2
2	85.	What type of conversion take place in Wittig reactions? Hence, write Wittig reagent.	2
2	86.	What are the reactants used in Wittig reactions? Write one ionic liquid used to carry out this reaction.	2
2	87.	What is a Wittig reaction? Write one ionic liquid used in this reaction	2
2	88.	What is main advantage of ionic liquids in Wittig reactions?	2
2	89.	What is PTC? Give an example	2
2	90.	Write any two examples of Phase transfer catalyst	2
2	91.	Give an example of PTC of ammonium salt.	2
2	92.	Give any two example of PTC of Phosphonium salt.	2
2	93.	Write the structure of 18-Crown-6-potassium cation.	2
2	94.	Write the structure of “Purple Benzene”	2
2	95.	How octyl chloride react with sodium cyanide in absence and presence of PTC?	2
2	96.	What is principal of Phase transfer catalyst?	2
2	97.	Write the equation for the synthesis of Triethyl benzyl ammonium Chloride.	2
2	98.	The use of visible light in Friedel-Crafts reaction is predominant. Justify	2
2	99.	What is use a photochemical alternative Friedel-Crafts reaction?	2
2	100.	Why George developed an alternative to the traditional Friedel-Crafts reaction?	2
2	101.	Name any two pharmacological drug produce using Friedel-Crafts reaction	2
2	102.	What is Photo Friedel-crafts acylation?	2
2	103.	Dimethyl carbonate consider as a green reagent. Give reasons	2
2	104.	Why dimethyl carbonate acts as environmentally innocuous material?	2

2	105.	How dimethyl carbonate acts as efficient methylating agent?	2
2	106.	A methylating agent dimethyl carbonate has advantage over methyl iodide and Dimethyl sulphate. Justify	2
2	107.	Why dimethyl carbonate is an efficient methylating agent then methyl iodide?	2
2	108.	Name any two green oxidant used in oxidation reaction.	2
2	109.	Give any two oxidainzing agent which is non toxic in nature	2
2	110.	What are green oxidants? Give an example	2
2	111.	Write one oxidation reaction with the use of green oxidant.	2
2	112.	What is the use of green oxidant?	2
2	113.	What is a supercritical fluid? Give an example	2
2	114.	What is super-critical carbon dioxide?	2
2	115.	What is supercritical carbon dioxide? Hence, write the critical temperature and critical pressure of CO_2.	2
2	116.	What is triple point? Write the phase diagram of carbon dioxide.	2
2	117.	Write the phase diagram of carbon dioxide and mark the triple point.	2
2	118.	What is critical temperature and critical pressure?	2
2	119.	Define critical temperature, critical pressure and triple point.	2
2	120.	Write any two properties of supercritical fluid.	2
2	121.	Why only CO_2 as the superficial fluids?	2
1	122.	Explain the green route for synthesis of Adipic acid. List out the disadvantages existed in Preceding Protocol.	4
1	123.	Discuss the Green Chemistry principle number 08 and 10 with suitable examples	4
1	124.	Calculate the atom economy of dehydrohalogenation of 2-bromo2-methyl propane.	4
1	125.	Mention any four greener stating materials.	4
1	126.	Differentiate the use of PNBS over use of NBS for bromination using an example.	4
1	127.	Write a note on green acid catalyst.	4
1	128.	Explain the advantages of biocatalytic conversions in relevance to green chemistry using an example.	4
1	129.	Enlist the disadvantages of using traditional solvents.	4
1	130.	Explicate the disadvantages of solvents.	4
1	131.	Illustrate aqueous phase Diels Alder Reaction.	4
1	132.	Write a note on microwave assisted Neat Reactions.	4

1	133.	Write any two synthesis using Furfural as starting material.	4
1	134.	Write any two synthesis using Glucose as starting material.	4
1	135.	Write any two application using Polymer supported reagents.	4
1	136.	What are Phase transfer catalyst? Explain its application.	4
1	137.	Solvents are important in Organic synthesis- Justify.	4
1	138.	Enlist the advantages of microwave heating.	4
1	139.	Write any two microwave assisted oxidation reaction	4
1	140.	Write a note on mw assisted Ugi reaction.	4
2	141.	What happen to the properties of ionic liquids if replacement of anion take place. Explain with an example	4
2	142.	Explain the synthesis of ionic liquids using quaternization step. Hence also write physical properties of ionic liquids	4
2	143.	Discus the synthesis of ionic liquids by Metathesis reaction using HPF_6 and KPF_6	4
2	144.	Write any five application of Ionic liquids	4
2	145.	Summarize a report on synthesis of ionic liquids	4
2	146.	Explain the procedure for the synthesis of pyridinium based ionic liquids	4
2	147.	Write the structure of i) [Bmim]BF_4^- ii) [NBPy]$AlCl_4^-$ iii) [Emim]$AlCl_4^-$ iv) [EtPy]CF_3COO^-	4
2	148.	Summarize a note on Alkyaltion process in presence of Ionic liquids	4
2	149.	Write a note on Alkylation process with use of ionic liquid like $[EtPy]^+[CF3COO]^-$. Hence, also write its structure.	4
2	150.	Write a note on application of ionic liquids in alkylation reaction.	4
2	151.	Explain hydroformylation process using Ionic liquids	4
2	152.	Draw conclusion on the hydroformylation process using Ionic liquids.	4
2	153.	Discus the investigation of Song et al., on hyroformylation reaction using ionic liquids.	4
2	154.	Summarize a report on epoxidation using Ionic liquids.	4
2	155.	Explain the synthesis of cylic ether (Epoxide) using Ionic liquids.	4
2	156.	Write a note on the synthesis of epoxide using a ionic liquids.	4

2	157.	Explain the application of ionic liquids in Friedel-Crafts alkylation.	4
2	158.	Summarize a note on Friedel-Crafts alkyaltion process in presence of Ionic liquids	4
2	159.	Write a note on Friedel-Crafts alkylation process with use of ionic liquid like $[EtPy]^+[CF3COO]^-$.	4
2	160.	Explain the application of ionic liquids in Diels-Alder reactions.	4
2	161.	Summarize a report on application of ionic liquids in Diels-Alder reactions	4
2	162.	Write a note on Diels-Alder reactions with use of ionic liquid.	4
2	163.	Discuss the importance of ionic liquids in Diels-Alder reactions.	4
2	164.	Explain the advantage of ionic liquids in Knoevengal condensation reaction.	4
2	165.	Summarize a report on Knoevengal condensation reaction in presence of Ionic liquids.	4
2	166.	Disuse the use of Ionic liquids in Knoevengal condensation reaction with example.	4
2	167.	Write a brief note on application of ionic liquids in Knoevengal condensation reaction.	4
2	168.	Explain Wittig reactions with use of ionic liquids.	4
2	169.	Describe the synthesis of olefin by witting reaction in presence of ionic liquids?	4
2	170.	Comparatively write the witting reaction in absence and presence of Ionic liquids.	4
2	171.	Summarize a report on the application of ionic liquids in witting reaction.	4
2	172.	Write any four advantage of PTC	4
2	173.	Summarize a report on application of Phase transfer catalyst	4
2	174.	Discus the mechanism of Norcarane synthesis using PTC.	4
2	175.	Write a note on the synthesis of 7,7dicholronorcarane under PTC condition. Hence, also write structure of “Purple Benzene”.	4
2	176.	How octyl chloride react with sodium cyanide in absence and presence of PTC? Explain	4
2	177.	Write a note on Phase transfer catalyst and its application.	4
2	178.	Summarize a report on Photo Friedel-crafts acylation	4
2	179.	Write a note on a photochemical alternative to Friedel-crafts reactions in the	4

		synthesis of Diazepam drug.	
2	180.	Describe the application of photochemical alternative to Friedel-crafts reactions for the synthesis of Doxepin drug.	4
2	181.	Explain a photochemical alternative to Friedel-crafts reactions for the synthesis of benzophenone.	4
2	182.	Write any two methylating agent which is toxic in nature.	4
2	183.	Discuss the properties of methylating agent dimethyl carbonate.	4
2	184.	Write a note on properties of dimethyl carbonate.	4
2	185.	Write the comparison of toxic and non toxicmethylating agent.	4
2	186.	Summarize a report on mono-C-methylation of aryl acetonitriles using dimethyl carbonate.	4
2	187.	Write a note application of green oxidant.	4
2	188.	Summarize a report on the application of green oxidant.	4
2	189.	Explain the application of green oxidant in oxidation reaction.	4
2	190.	Write any four oxidation reaction in which green oxidant is used as oxidizing agent.	4
2	191.	What is super-critical carbon dioxide? Write the phase diagram of carbon dioxide and mark the triple point.	4
2	192.	Explain homogeneous catalytic hydrogenation of supercritical carbon dioxide.	4
2	193.	What is triple point? Explain homogeneous catalytic asymmetric hydrogenation reaction in $scCO_2$	4
2	194.	What is critical pressure and critical temperature. Hence write the production of dimethyl sulfoxide using $scCO_2$.	4
2	195.	Write a note on application of supercritical carbon dioxide in hydrogenation reaction.	4
1	196.	Water is a green solvent. Justify by mentioning its applications.	5
1	197.	Explain Green chemistry Principle 10 (design for degradation).	5
1	198.	Clarify how principle “Reduce Derivatives” can be achieved.	5
1	199.	Illustrate the application of green principle “Designing safer chemicals”.	5
1	200.	How do you implement green principle 3- ‘Less Hazardous Chemical Synthesis”	5
1	201.	Furfural is a green starting material. Justify.	5
1	202.	Explain the synthesis and application of Polylactic acid.	5
1	203.	What are Polymer supported Reagents? Explain.	5
1	204.	What are Biocatalyst? Explain.	5

1	205.	Illustrate Phase transfer catalyst.	5
1	206.	Illustrate the application water as green solvent.	5
1	207.	Illustrate microwave assisted Functional Group Transformation reactions.	5
1	208.	Illustrate mw assisted condensation reactions.	5
1	209.	Elaborate on Multicomponent Reactions assisted by microwave.	5
1	210.	Explain the mechanism of microwave heating.	5
1	211.	a. Maleic anhydride may be prepared using two routes: i.  **Maleic Anhydride** ii  **Maleic Anhydride** By considering the green metrics (atom economy, with calculation), which route would you recommend to industry?	6
1	212.	Glucose is greener starting material. Justify.	6
1	213.	Addition Reactions are more efficient than Substitution reaction. Justify by calculating atom economy for afore mentioned reactions.	6
1	214.	Mention the Anastas' twelve principles of green chemistry.	6
1	215.	Write a note Aqueous Phase Reactions.	6
1	216.	Explain the functions of the major classes of enzymes.	6
1	217.	Write a note on i. Microwave assisted Neat Reactions. ii. Microwave assisted solid support reactions.	6
1	218.	Write an elaborate note on Green principle-1	6
1	219.	Elaborate on Green catalyst.	6
1	220.	Write elaborately on green starting material.	6
1	221.	Elaborate on Biocatalyst.	6
1	222.	Write a note on a. Acid Green catalyst b. Biocatalyst	6
1	223.	Write a note on a. Oxidation Green catalyst b. PTC	6
1	224.	Write a comparison note between conventional and microwave heating.	6

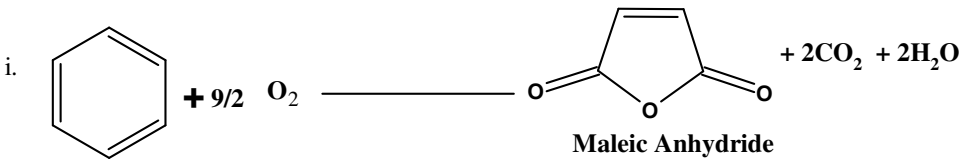
1	225.	Elaborate on mw assisted oxidation and reduction reactions	6
2	226.	Explain the synthesis of ionic liquids using quaternization step. Hence also write physical properties of ionic liquids. Hence, write the structure of $[EtPy]CF_3COO^-$.	6
2	227.	Why ionic liquids are so attractive? Discus the synthesis of ionic liquids by Metathesis reaction using HPF_6 and KPF_6	6
2	228.	Why ionic liquids are so attractive? Write any five application of Ionic liquids	6
2	229.	Summarize a report on synthesis of ionic liquids	6
2	230.	What are ionic liquids? Explain the procedure for the synthesis of pyridinium based ionic liquids.	6
2	231.	What are ionic liquids? Write the structure of i) $[Bmim]BF_4^-$ ii) $[Bmim]PF_6^-$ iii) $[NBPy]AlCl_4^-$ iv) $[Emim]AlCl_4^-$ v) $[EtPy]CF_3COO^-$	6
2	232.	What is alkylation? Name one ionic liquids used in this reaction and hence, Summarize a note on Alkyaltion process in presence of Ionic liquids	6
2	233.	What are ionic liquids? Write a note on Alkylation process with use of ionic liquid like $[EtPy]^+[CF3COO]^-$. Hence, also write its structure.	6
2	234.	Write a note on application of ionic liquids in alkylation reaction. Hence, name the pyridinium ionic liquid used in alkylation process.	6
2	235.	Explain Hydroformylation process using Ionic liquids. Hence write the structure of mixed ionic liquid used in	6
2	236.	What is Hydroformylation reaction? Draw conclusion on the hydroformylation process using Ionic liquids.	6
2	237.	Discus the investigation of Song et al., on hyroformylation reaction using ionic liquids.	6
2	238.	Summarize a report on application of ionic liquids in hyroformylation reaction	6
2	239.	Write the structure of $[Bmim]BF_4$. Hence explain epoxidation reaction using ionic liquids.	6
2	240.	Explain the synthesis of cylic ether (Epoxide) using Ionic liquids.	6
2	241.	Write a note on the synthesis of epoxide using a ionic liquids with its application	6
2	242.	What is Friedel-Crafts alkylation? Name one ionic liquids used in this reaction and hence, Summarize a note on Friedel-Crafts alkylation in presence of Ionic liquids.	6

2	243.	What are ionic liquids? Write a note on Alkylation process with use of ionic liquid like $[EtPy]^+[CF3COO]^-$. Hence, also write its structure.	6
2	244.	Write a note on application of ionic liquids in Friedel-Crafts alkylation reaction. Hence, name any two pyridinium ionic liquid used in this process.	6
2	245.	What are ionic liquids? Write report on application of ionic liquids in Friedel-Crafts alkylation reaction.	6
2	246.	Explain the application of ionic liquids in Diels-Alder reactions. Hence write the structure of [bmim][OTf] and[bmim][BF_4].	6
2	247.	Summarize a report on application of ionic liquids in Diels-Alder reactions. Hence write the structure of [bmim][OTf] and[bmim][BF_4].	6
2	248.	Write a note on Diels-Alder reactions with use of ionic liquid. Hence write the structure of [bmim][OTf] and[bmim][PF_6].	6
2	249.	Discuss the importance of ionic liquids inDiels-Alder reactions. Hence write the structure of [bmim][OTf] and[bmim][PF_6].	6
2	250.	Explain the advantage of ionic liquids in Knoevengal condensation reaction. Hence, write the structure of (1-((4-Cholrophenyl)amine)-1-oxo-propan-2-aminium per chlorate.	6
2	251.	Why Ionic liquids is used in Knoevengal condensation reaction? Summarize a report on Knoevengal condensation reaction in presence of Ionic liquids.	6
2	252.	Disuse the use of Ionic liquids in Knoevengal condensation reaction with example. Hence, write the structure of (1-((4-Cholrophenyl)amine)-1-oxo-propan-2-aminium per chlorate.	6
2	253.	Write a note on application of ionic liquids in Knoevengal condensation reaction.	6
2	254.	Explain Wittig reactions with use of ionic liquids. What is main advantage of ionic liquids in Wittig reactions?	6
2	255.	What is main advantage of ionic liquids in Wittig reactions? Describe the synthesis of olefin by witting reaction in presence of ionic liquids?	6
2	256.	Comparatively write the witting reaction in absence and presence of Ionic liquids. Hence, write the structure of two ionic liquids used in this process.	6
2	257.	Summarize a report on the application of ionic liquids in Witting reaction. Hence, write the structure of two ionic liquids used in this process.	6
2	258.	Summarize a report on application of Phase transfer catalyst. Hence, Write the structure of “Purple Benzene”	6
2	259.	Discus the mechanism of Norcarane synthesis using PTC. Hence, Write the structure of “Purple Benzene”	6
2	260.	Write a note on the synthesis of 7,7dicholronorcarane under PTC condition. Hence, also write structure of “Purple Benzene”.	6
2	261.	How octyl chloride react with sodium cyanide in absence and presence of PTC? Explain. Also, write any two example of PTC of Phosphonium salt.	6
2	262.	Why George developed an alternative to the traditional Friedel-Crafts reaction?	6

		Hence, Summarize a report on PhotoFriedel-crafts acylation.	
2	263.	What is Photo Friedel-crafts acylation? Write a note on a photochemical alternative to Friedel-crafts reactions in the synthesis of Diazepam drug.	6
2	264.	What is Photo Friedel-crafts acylation? Describe the application of photochemical alternative to Friedel-crafts reactions for the synthesis of Doxepin drug.	6
2	265.	Why George developed an alternative to the traditional Friedel-Crafts reaction? Hence, explain a photochemical alternative to Friedel-crafts reactions for the synthesis of benzophenone.	6
2	266.	What is Photo Friedel-crafts acylation? Explain the use of photochemical alternative to Friedel-crafts reactions.	6
2	267.	Explain the methylation process using DMC Under continuous flow GL-PTC.	6
2	268.	Discuss the mechanism of methylation process with DMC using gas-liquid phase transfer catalysis.	6
2	269.	Write a report on mono-C-methylation of arylacetonitriles using dimethyl carbonate.	6
2	270.	Describe a note on Mechanism of the mono-C-methylation of CH_2-active compounds with DMC	6
2	271.	Write a note on dimethyl carbonate acts a methylating agent.	6
2	272.	Write a note application of green oxidant with an example.	6
2	273.	What is green oxidant? Summarize a report on the application of green oxidant.	6
2	274.	What is green oxidant? Explain the application of green oxidant in oxidation reaction.	6
2	275.	What is green oxidant? Write any five oxidation reaction in which green oxidant is used as oxidizing agent	6
2	276.	Summarize a report on use supercritical carbon dioxide in synthetic chemistry.	6
2	277.	Discuss the use of supercritical carbon dioxide in homogeneous catalytic hydrogenation reaction.	6
2	278.	Write a note on production of DMF and homogeneous catalytic hydrogenation using $scCO_2$.	6

Question Paper Pattern- Model Question Paper

Q.P. Code:MCH20A311

St. Philomena's College (Autonomous), Mysore			
I Semester –Course M.Sc Final Examination			
Subject : Chemistry			
Title: Green Chemistry (SC)			
Time: 3 hrs			**Max Marks: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Differentiate Percentage yield from Atom economy.	2
	b	Mention any four green reagents.	2
	c	What are biocatalyst? Give an example.	2
	d	Why microwave heating is described as an "**instant on- instant off"?**	2
	e	Write green synthesis protocol for synthesis of Adipic acid.	2
	f	Give an example for mw assisted oxidation reaction.	2
	g	Why molten NaCl does not consider as ionic liquids?	2
	h	Write the structure of 1-Butyl-3-methylimidazolium hexafluorophophate	2
	i	Write the structure of Katsuki-type catalyst.	2
	j	Ionic liquids of amino acid amides used in Knoevengal condensation reaction. Give reasons.	2
	k	What is PTC? Give an example.	2
	l	A methylating agent dimethyl carbonate has advantage over methyl iodide and dimethyl sulphate. Justify	2
PART-B			
		Answer any Five of the following:	**5 x 10= 50**
2	a	a. Maleic anhydride may be prepared using two routes: i.  **+ 9/2 O_2 ——— + $2CO_2$ + $2H_2O$** **Maleic Anhydride** ii  **+ 3 O_2 ——— + $3H_2O$** **Maleic Anhydride** By considering the green metrics (atom economy, with calculation), which route would you recommend to industry?	6
	b	Calculate the atom economy of dehydrohalogenation of 2-bromo2-methyl propane.	4

3	a	Illustrate the application of green principle “Designing safer chemicals”.	5
	b	Illustrate microwave assisted Functional Group Transformation reactions.	5
4	a	Write a note on a. Acid Green catalyst b. Biocatalyst	6
	b	Write a note on microwave assisted Neat Reactions.	4
5	a	Write a comparison note between conventional and microwave heating	6
	b	Explain the procedure for the synthesis of pyridinium based ionic liquids	4
6	a	Explain the synthesis of cylic ether (Epoxide) using Ionic liquids.	6
	b	Discus the investigation of Song et al., on hyroformylation reaction using ionic liquids.	4
7	a	Summarize a report on use supercritical carbon dioxide in synthetic chemistry.	6
	b	Describe the synthesis of olefin by witting reaction in presence of ionic liquids?	4
8	a	Why George developed an alternative to the traditional Friedel-Crafts reaction? Hence, explain a photochemical alternative to Friedel-crafts reactions for the synthesis of benzophenone.	6
	b	Summarize a report on mono-C-methylation of aryl acetonitriles using dimethyl carbonate.	4